

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B. Tech (CBCS) Degree Examinations

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December 2025

CC3101-1 – CRYPTOGRAPHY AND NETWORK SECURITY

(For CCE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- 1. A transposition cipher reorders (permutes) symbols in a**
A) block of packets
B) block of slots
C) block of signals
D) block of symbols
- 2. Which is not an objective of network security?**
A) identification
B) authentication
C) access control
D) lock
- 3. The man-in-the-middle attack can endanger the security of the Diffie Hellman if two parties are not**
A) authenticated
B) joined
C) submit
D) separate
- 4. In cryptography, the order of the letters in a message is rearranged by**
A) transpositional ciphers
B) substitution ciphers
C) Semi quadratic ciphers
D) quadratic ciphers
- 5. Cryptanalysis is used**
A) to find some insecurity in a cryptographic scheme
B) to increase the speed
C) to encrypt the data
D) to make new ciphers
- 6. In public key encryption system if A encrypts a message using his private key and sends it to B**
A) if B knows it is from A he can decrypt it using A's public key
B) Even if B knows who sent the message it cannot be decrypted
C) It cannot be decrypted at all as no one knows A's private key
D) A should send his public key with the message
- 7. What happens to your data when it is encrypted?**
A) It is transferred to a third party, encoded, then sent back.
B) It is compressed, renamed, and archived.
C) It is sent through a series of supercomputers to be compressed multiple times.
D) It is recorded to retain privacy from third parties.
- 8. How to avoid Man-in-the-middle attacks?**
A) Accept every SSL certificate, even the broken ones
B) Use connections without SSL
C) use HTTPS connections and verify the SSL certificate
D) Use connectionless without SSL
- 9. AES uses a _____ bit block size and a key size of _____ bits.**
A) 32; 64, 128 or 256
B) 64; 128 or 192
C) 126; 128, 192, or 256
D) 128; 128, 192, or 256

10. What is the size of the key in the SDES algorithm?
 A) 32 bits B) 22 bits
 C) 16 bits D) 10 bits
11. Why is the distribution of keys considering a problem in symmetric-key cryptography?
 A) It requires frequent updates to the keys B) It is difficult to ensure secure and timely delivery of keys to all parties
 C) It relies on a single point of failure for key management D) It becomes more vulnerable as the number of parties increases
12. What is the primary purpose of SSL/TLS in website security?
 A) To store user data securely B) To encrypt data transmission between the server and client
 C) To improve website speed D) To block malware
13. What is the main risk of not validating user input on a website?
 A) Poor website performance B) Vulnerability to SQL Injection
 C) Increased loading time D) Incorrect analytics data
14. What is a "man-in-the-middle" attack?
 A) An attacker intercepting communications between two parties B) A physical theft of a server
 C) A denial-of-service attack D) Unauthorized login attempts
15. WPA2 is used for security in _____.
 A) Ethernet B) Bluetooth
 C) Wi-Fi D) Email
16. _____ is the protocol designed to create security associations, both inbound and outbound.
 A) SA B) CA
 C) KDC D) IKE
17. _____ is actually an IETF version of _____.
 A) TLS; TSS B) SSL; TLS
 C) TLS; SSL D) SSL; SLT
18. The full form of SSL is
 A) Serial Session Layer B) Secure Socket Layer
 C) Session Secure Layer D) Series Socket Layer
19. What is the primary purpose of Secure Shell (SSH)?
 A) To transfer files between computers insecurely. B) To provide a secure channel over an unsecured network for remote command-line login and execution.
 C) To manage local files on a single computer. D) To browse websites anonymously.
20. What is the primary goal of encryption in web applications?
 A) To compress data B) To increase data integrity
 C) To secure data transmission and storage D) To improve server performance

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is the Caesar Cipher? Explain its encryption and decryption processes with an example.	08	L*2	1	1
	b) What is the Playfair Cipher? Explain its encryption and decryption process with an example.	08	L3	1	1, 2
2.	a) Explain the working of the Data Encryption Standard (DES). Analyze its encryption process, key schedule, and strengths and weaknesses.	08	L3	1	2
	b) Explain the structure, key sizes, and basic working principle of Advanced Encryption Standard.	08	L2	2	1, 2

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|-------|---|----|----|---|---|
| 3. a) | Analyze the principles of public-key cryptosystems. Explain how these principles provide confidentiality, authentication, and non-repudiation, with examples. | 08 | L3 | 2 | 2 |
| b) | Explain the RSA algorithm. Analyze its key generation, encryption, and decryption processes, and discuss its strengths and limitations. | 08 | L2 | 2 | 2 |

Unit – II

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|-------|---|----|----|---|------|
| 4. a) | Explain the Man-in-the-Middle (MitM) attack in Diffie-Hellman key exchange. Analyze how the attack occurs and discuss possible countermeasures. | 08 | L4 | 3 | 1, 2 |
| b) | What are key exchange protocols? Explain their purpose and working. | 04 | L2 | 3 | 1 |
| c) | What is Elliptic Curve Cryptography (ECC)? Explain its basic concept and key advantages. | 04 | L2 | 3 | 1 |
| 5. a) | Explain hierarchical key control in key management and distribution. Discuss its structure, working, and advantages. | 08 | L2 | 3 | 1, 2 |
| b) | Explain remote user authentication using asymmetric encryption. Discuss its working, components, and advantages. | 08 | L2 | 4 | 1, 2 |
| 6. a) | Explain identity management for user authentication. Discuss its components, working, and advantages in secure systems. | 08 | L2 | 4 | 1, 2 |
| b) | Explain identity management for user authentication. Discuss its components, working, and advantages in secure systems. | 08 | L2 | 4 | 1, 2 |

Unit – III

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|-------|---|----|----|---|---|
| 7. a) | Illustrate the types of security threats faced while using the Web. | 08 | L2 | 5 | 1 |
| b) | Explain SSH Protocol Stack. Describe its architecture, layers, and their functions. | 08 | L2 | 5 | 1 |
| 8. a) | What are the main applications of IP Security (IPSec)? Explain briefly. | 08 | L2 | 5 | 1 |
| b) | Illustrate the various benefits of IPSec. | 08 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
